

Rahul Kumar

📍 Bokaro | ✉ rahulkumaraec.cse@gmail.com | 🌐 Portfolio | 🔗 LinkedIn | 🐙 GitHub

Summary

Full Stack Engineer (B.Tech CSE '27) with production internship experience in **React/Node.js**, **GraphQL**, and **AI-integrated** systems. Built real-world platforms with LLM APIs, WebSocket-based real-time features, and code execution engines 2,000+ DSA problems solved. Targeting SDE internships and full-stack roles at product-focused companies.

Education

Asansol Engineering College

Aug 2023 – July 2027

Bachelor of Technology in Computer Science

Relevant Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP, Software Engineering

Experience

Frontend Engineer Intern

Bangalore, India

Advertyzement · React · GraphQL · TypeScript · Vitest

July 2025 – Jan 2026

- Engineered **GraphQL** integration replacing REST in a 3-engineer React frontend, eliminating redundant over-fetching and cutting network calls by **50%** while improving data retrieval speed by **25%**.
- Architected a modular test suite with **Vitest** and **React Testing Library**, covering critical user flows (auth, form submission, API states) and enforcing validation on every pull request.
- Executed a mobile-first, cross-browser UI overhaul (Chrome, Firefox, Safari), achieving **100% Lighthouse accessibility score** and consistent sub-2s LCP across all device classes.
- Migrated 8,000+ lines of legacy JavaScript to **ES Modules** with tree-shaking, reducing bundle size by **20%** and eliminating circular dependency tech debt.

Projects

LoopMock — AI-Powered Interview Preparation Platform

GitHub | Live

React · Vite · Zustand · Node.js · Express · MongoDB · Groq LLaMA-3 · Judge0 CE · Socket.io

- Engineered full-stack interview platform (DSA, system design, behavioral, coding) with **Groq LLaMA-3** generating questions and scoring answers on correctness, clarity, depth, and communication in **under 3 seconds** per submission.
- Built a **LeetCode-style code judge** integrating **Judge0 CE** (60+ languages) with hidden test cases and strict output comparison; abstracted the execution layer behind a service interface, enabling zero-downtime migration when Piston API shut down.
- Implemented **real-time peer mock interviews** via **Socket.io** rooms with queue-based matchmaking and role assignment; secured with **JWT + httpOnly refresh tokens**, Google OAuth (Passport.js), and tiered rate limiting (auth/API/AI).
- Designed an **AI resume analyzer** that parses uploaded PDFs, validates content before the AI call (rejecting non-resumes), and returns skill gap analysis with role-specific interview questions via structured Groq prompt engineering.

SecondBrain — Personal Knowledge Management Platform

GitHub

TypeScript · React · Tailwind CSS · Node.js · Express · MongoDB · JWT

- Architected a full-stack content aggregation platform (YouTube, Twitter, notes) with **indexed MongoDB queries** and server-side caching, cutting content retrieval latency by **50%** under concurrent load.
- Implemented **JWT** auth with refresh token rotation and **bcrypt** hashing, enforcing per-user data isolation; optimised React bundle with lazy loading and Tailwind purging, achieving **40% faster** initial page load.

Technical Skills

- Languages:** JavaScript (ES2022+), TypeScript, C++, C
- Frontend:** React.js, Next.js, Tailwind CSS, Zustand, Redux, Monaco Editor, HTML5, CSS3
- Backend:** Node.js, Express.js, REST APIs, GraphQL, Socket.io, JWT, Passport.js, node-cron
- Databases & AI:** MongoDB, PostgreSQL, MySQL, Mongoose, Redis, Groq API, OpenAI API, LLM/RAG (learning)
- DevOps & Tools:** Git, GitHub, Vercel, Render, Railway, Postman, Judge0 CE, Vitest, Linux

Achievements

- Solved **2,000+** DSA problems across LeetCode, GeeksforGeeks, and Codeforces top **5%** globally on LeetCode.
- Ranked **831st globally** in LeetCode Biweekly Contest 161 out of 30,000+ participants.
- Secured **77th rank** in the DSA MasterMind round of CodeClash Hackathon among 5,000+ participants.